



JAIPUR NATIONAL UNIVERSITY

Semester/Year End Examination, May-2024

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. Biomedical) /B.Tech. (Chemical) /B.Tech. (Civill) / B.Tech.(Mechanical) /B.Tech. (CSE) / B.Tech. (CSE- AIML)/B.Tech. (CSE Cyber Security) / B.Tech. (Electrical)
Semester/Year:	2
Paper Code:	BTEMACO204T22
Title of Paper:	Engineering Mathematics II
Time: 3 hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

This question paper consists of two sections, Section A and Section B. Section-A consists of seven questions and Section-B consists of six question. Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question.

Question
Number

Marks
Allotted

SECTION-A

- Q 1. Show that the $\bar{a} = [1 \ 1 \ 1]$, $\bar{b} = [1 \ 2 \ 0]$, and $\bar{c} = [0 \ 1 \ 1]$ are linearly independent. 4
- Q 2. Let $C = \begin{bmatrix} -5 & 2 \\ -7 & 4 \end{bmatrix}$. Find its eigenvalues and eigenvectors. 4
- Q 3. Determine the half range Fourier sine series for $f(x) = e^x$ in $0 < x < 1$. 4
- Q 4. Show that the differential equation $x \cos\left(\frac{y}{x}\right) \left(\frac{dy}{dx}\right) = y \cos\left(\frac{y}{x}\right) + x$ is homogenous and solve it. 4
- Q 5. Compute the first three harmonics of the Fourier series of 4

$f(x)$ given by the following table.

$x:$	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$
$f(x):$	1.0	1.4	1.9	1.7	1.5	1.2

- Q 6. Find the solution of $p + q = \cos x$. 4
- Q 7. Solve the 2nd order differential equation $9y'' + 12y' + 29y = 0$. 4

SECTION-B

- Q 1. Find the rank of given matrix after reduce the matrix into normal form 10

$$\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

- Q 2. Find the Fourier series for $y = x^2$ in $-\pi < x < \pi$. Using the two values of y , show that 10

$$\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \dots = \frac{\pi^4}{90}$$

- Q 3. Show that the matrix $A = \begin{bmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{bmatrix}$ is diagonalizable. 10

- Q 4. Solve $xy' + 3y = 4x^2 - 3x$. 10

- Q 5. Solve $z = px + qy + p^2 + q^2$ using Charpit's method. 10

- Q 6. Find the complete solution of the second order differential equation $y'' - 6y' + 5y = e^{-3x}$. 10